UNICITY

CUT OUT THE MIDDLEMAN.
MOVE THE VALUE FREELY.
PROVE WHO IS ON THE OTHER END.

Fair access takes all three. The first two were solved a decade ago. The third — a dollar that knows who may receive it — is what Unicity builds.

TALLINN · ZUG · ABU DHABI

THE HARD ONE: PROVING WHO IS ON THE OTHER END.

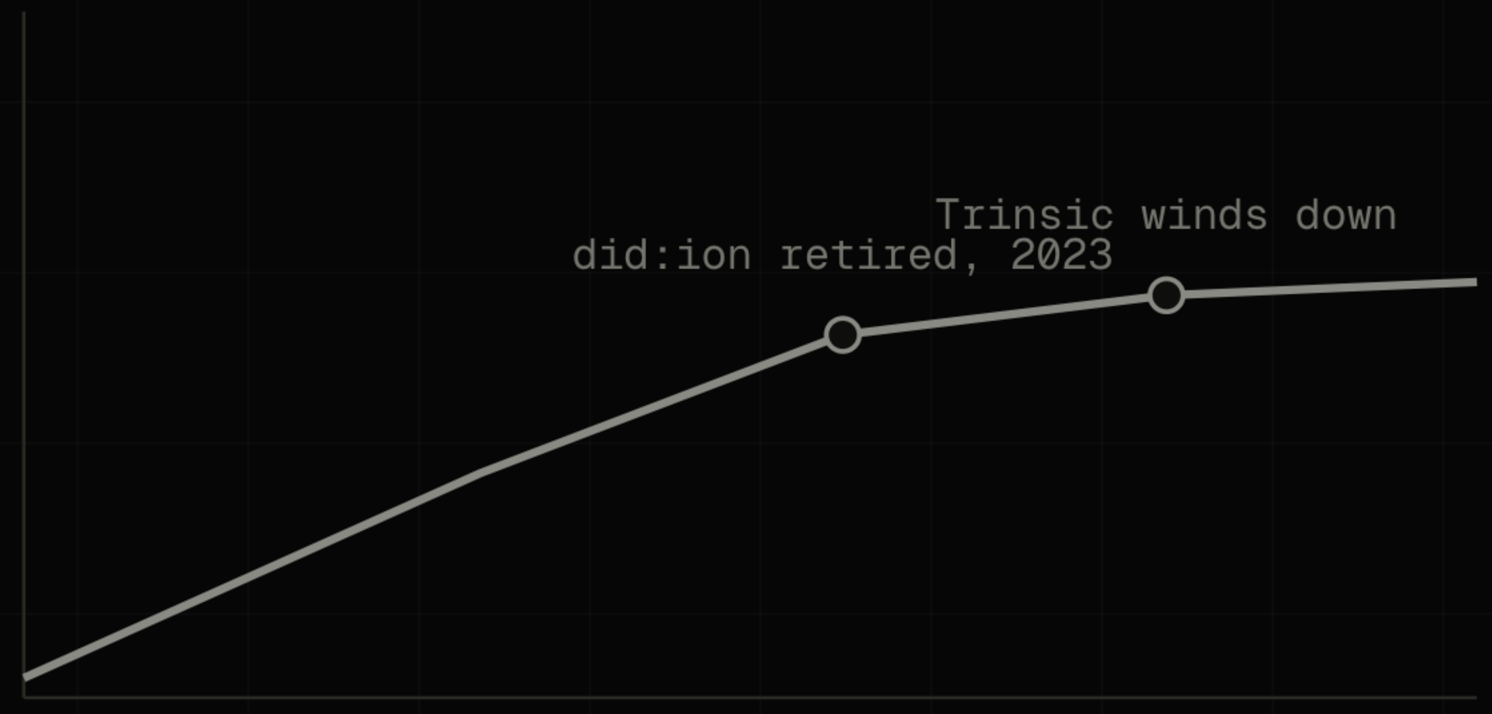
01	**CUT OUT THE MIDDLEMAN** A decade of open infrastructure settled it.	SOLVED
02	**LET VALUE MOVE FREELY** Stablecoins now move hundreds of billions a year.	SOLVED
03	**PROVE WHO IS ON THE OTHER END** So the money itself knows who may receive it.	STILL OPEN

The rest of this is how the money proves it — with the math, and the team behind it.

IDENTITY REACHED THE RIGHT ARCHITECTURE BEFORE THE MARKET COULD CARRY IT.

GlobaliD had it early; the field stalled at the same wall — did:ion retired in 2023, Trinsic wound down. The credential lived beside the transaction, and a check that can be skipped gets skipped.

USBC drew the right conclusion: put identity inside the dollar. Inside one charter it settles. Across strangers, it does not.



digital identity — climbs, then flattens

MOST OF THE TRAFFIC ONLINE IS ALREADY MACHINES – AND THEY HAVE STARTED PAYING EACH OTHER.

57.5%

OF WEB REQUESTS ARE AUTOMATED, NOT HUMAN — CLOUDFLARE, 2026

Those machines have begun settling value directly — roughly **100 million payments on Base in nine months** (Chainalysis), about thirty-one cents each. Legacy rails carry a fee floor above thirty cents, so sub-dollar machine payments are impossible on the old infrastructure.

> A machine cannot wait on hold while an intermediary clears it. The check has to travel with the money.

EVERY SHARED LEDGER DOES THE SAME FOUR THINGS FOR EVERYONE.

Proof-of-work, proof-of-stake, or DPoS — a shared ledger must broadcast, order, validate and record every transaction in global state. That requirement is the ceiling, and the privacy machines need (ZK, FHE) collapses it further. When billions of sub-cent agent events arrive, every node must process every one.

BROADCAST

Every node hears every transaction.

ORDER

Global agreement on the sequence of all of them.

VALIDATE

Every node re-runs every transaction.

RECORD

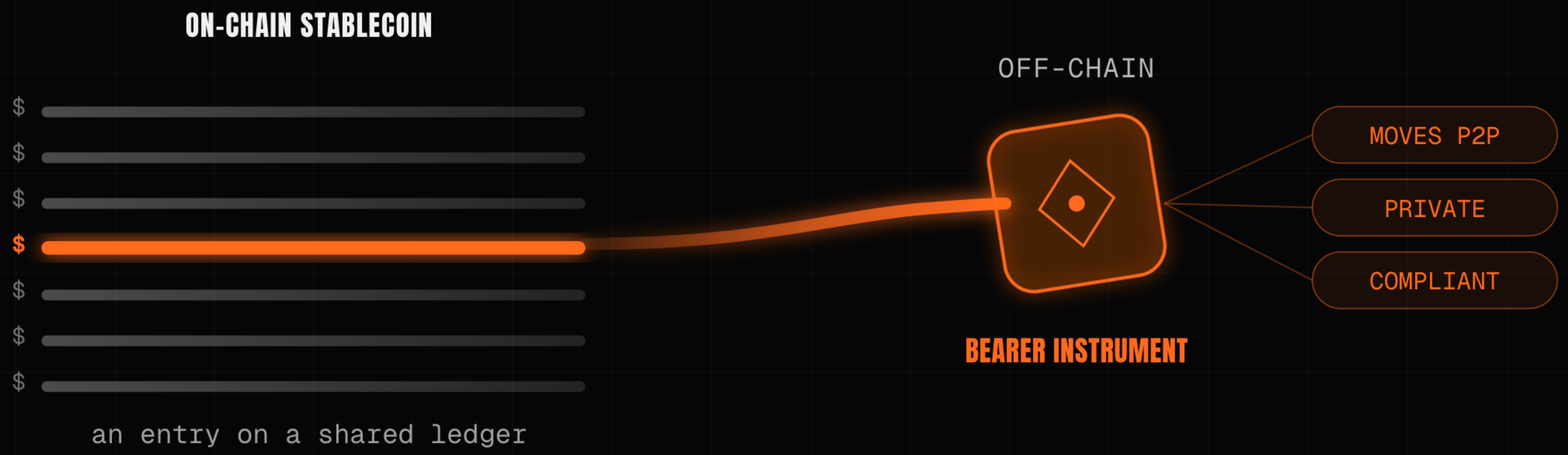
Every node stores the whole global state, forever.

> You cannot optimise your way out of this. The bottleneck is the design, not the implementation.

06 · HOW IT WORKS · OFF THE LEDGER

A DOLLAR ON-CHAIN IS A ROW IN A LEDGER SOMEONE ELSE KEEPS.

Unicity makes it a bearer file — a native data type, like a file or a process the system holds. It carries its own value and its own proof, moves directly between two parties the way cash changes hands, and the recipient verifies it on arrival against the token alone.



SELF-CONTAINED

The value lives in the file, not a row a network keeps for you.

SELF-PROVING

It carries its own proof of validity, minted against a locked source-chain asset.

VERIFIES ON ARRIVAL

The recipient checks it locally, offline, over any channel, with no chain to call.

07 · HOW IT WORKS · ONE QUESTION

THE ONLY JOB A CHAIN EVER HAD: HAS THIS BEEN SPENT?

The network receives an opaque commitment and answers one thing — spent or unspent — never the data, the counterparties, or the amount. It is the inevitable end of blockchain unbundling: the chain keeps the one job no one else can do, and everything else moves to the edge.

CONSENSUS & EMISSION

decentralized base security

↓ certified state root

UNIQUENESS ORACLE

stores only proofs that token states are unique

↓ inclusion proofs

EXECUTION LAYER

agents and users transact client-side, off-chain

OPAQUE COMMITMENT

The oracle sees only spent or unspent, never the trade itself.

PROVES THE TREE, NOT THE TRADE

A zero-knowledge proof attests the sparse-Merkle-tree state transition, not each payment.

THREE PROOF TYPES

Inclusion (registered, eternal), exclusion (low-latency settlement), unicity (the tree has not forked).

UNICITY PROVES UNIQUENESS. NOTHING ELSE.

Proving no double-spend is the oracle's only job. The three things a normal blockchain does for everyone — communication, storage, validation — move off-chain to the parties who actually care, handled with ordinary tools. That relocation, not a throughput number, is why it scales.

COMMUNICATION

The asset carries its own proof; the wire just moves bytes — NOSTR, email, QR.

STORAGE

State lives with the owner, not on every node — browser, device, IPFS.

VALIDATION

The recipient verifies; no validator set in the path — client-side, in parallel.

A named primitive, not a novelty: client-side validation and single-use seals (Bitcoin Optech, RGB, Nervos CKB).

THE INEVITABLE UNBUNDLING OF BLOCKCHAIN. WE TOOK IT TO ITS CONCLUSION.

Each generation of consensus removed work from the network.

2009 · BITCOIN

CORRECTNESS + ORDERING

Every node certifies every transaction and agrees on the order.

2023 · SUI · FASTPAY

CORRECTNESS ONLY

Global ordering removed for assets that share no state.

2026 · UNICITY

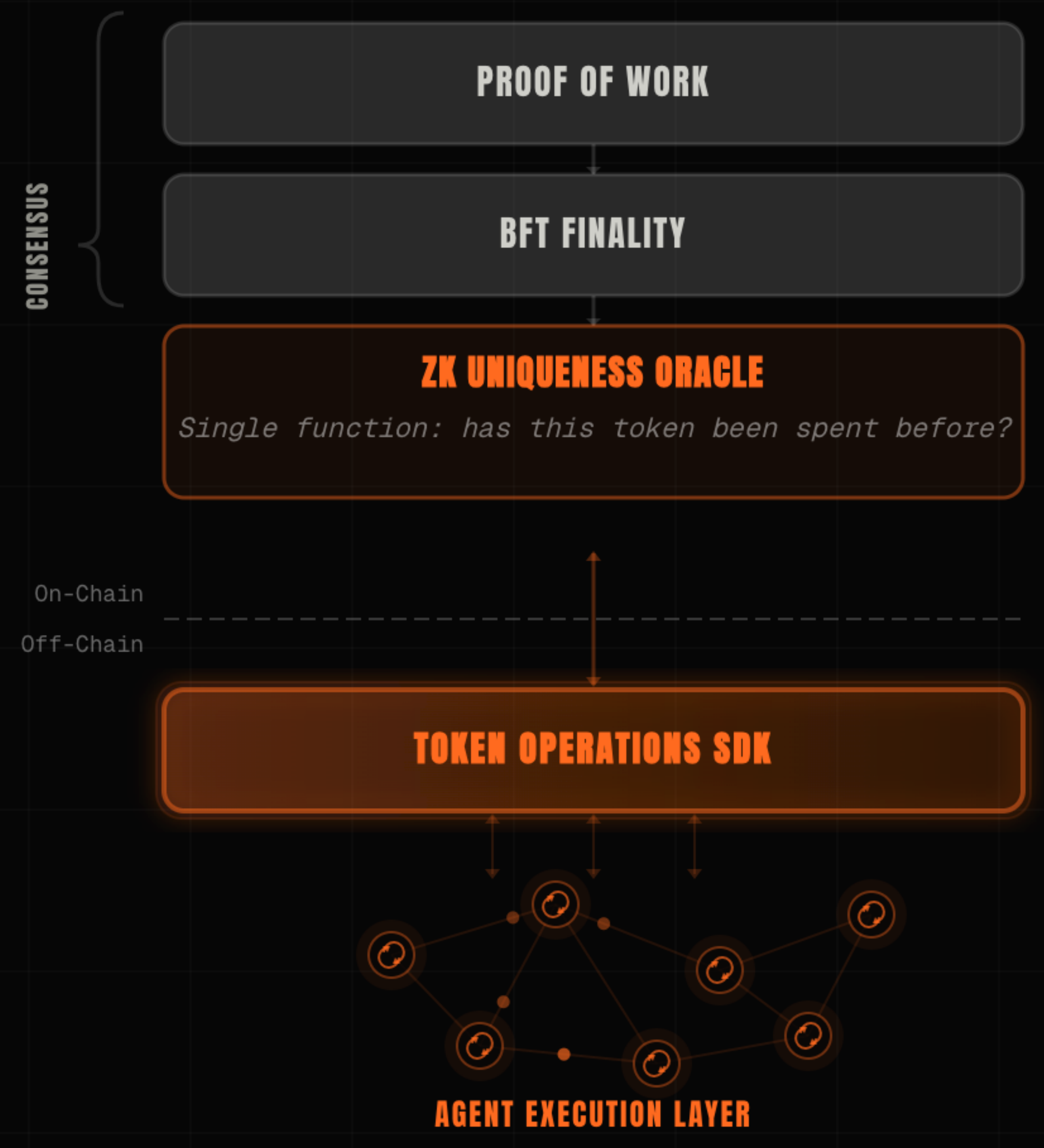
UNIQUENESS ONLY

The network attests one thing — has this been spent. Correctness moves to the edge.

> Not a competitor to chains — the logical endpoint of a decade of consensus research.

A MINIMAL CHAIN, AND AN ECONOMY THAT RUNS OFF IT.

The consensus layer stays deliberately small; everything a user experiences is off-chain, joined by a thin SDK. We use proof-of-work to replicate Bitcoin's trust model: trust no one. This is the engineering, not a roadmap.



ON-CHAIN, MINIMAL

RandomX proof-of-work, then chained HotStuff for ~1s finality, then the uniqueness oracle.

THE SDK

A thin bridge in JS, Java and Rust between off-chain agents and on-chain proofs.

AGENT EXECUTION LAYER

Transport (I2P, TCP/IP), messaging (NOSTR), storage, runtime — all peer-to-peer.

THE RULE LIVES INSIDE THE TOKEN.

Each token carries its own receive-condition — KYC, jurisdiction, sanctions — a more advanced version of Bitcoin's locking script. A transfer to a party that fails it cannot be constructed: no issuer to call, no monitor afterward. The issuer and verifier are configurable, and the token's whole history can be verified compliant.

PROGRAMMABLE

Signature, multi-sig, k-of-n threshold or time-lock, written once and travelling with the token.

PROTOCOL-ENFORCED

The check lives in the asset, not an app or custodian — no intermediary to bypass.

ISSUER-DEFINED, IN-BAND

The issuer sets who may hold it once; an unsatisfied transfer fails locally, with no callback.

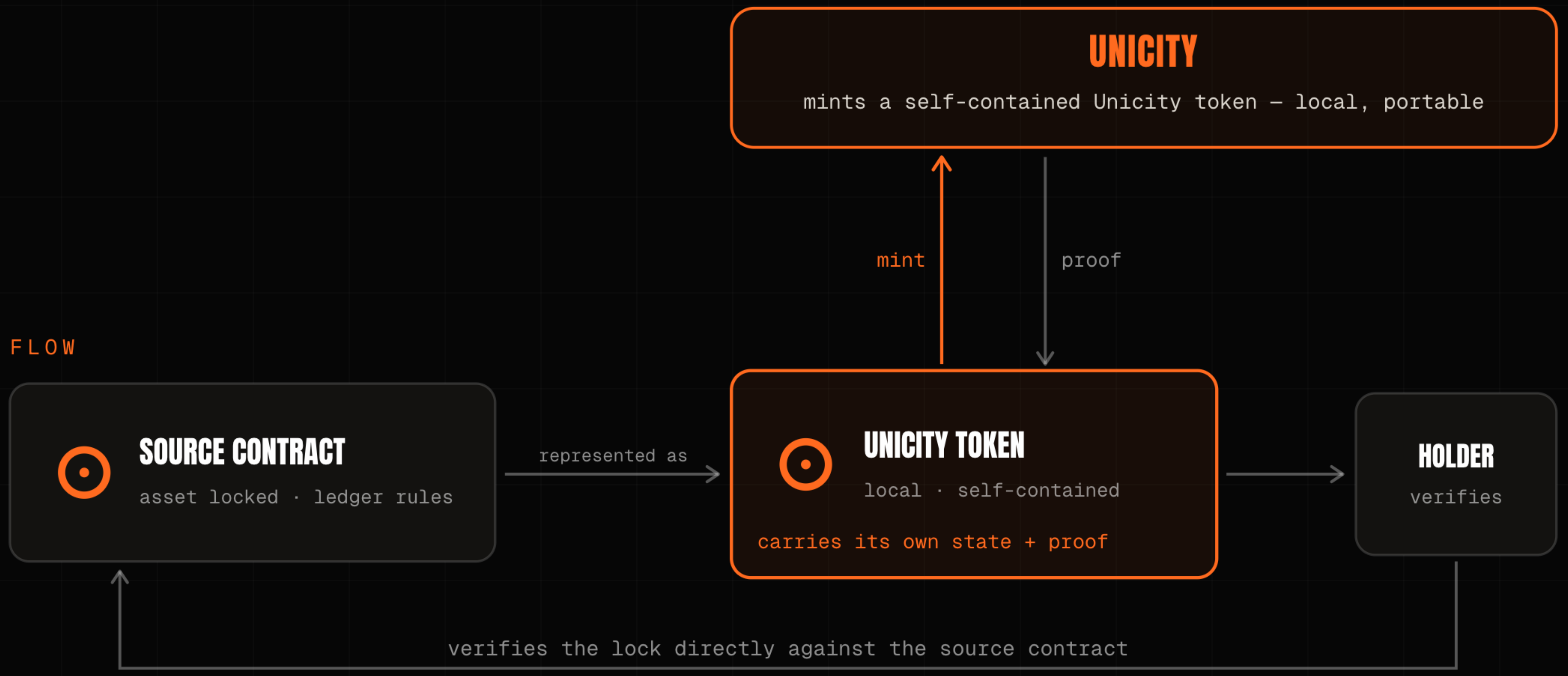
TOTAL PRIVACY BY DEFAULT. THREE OBSERVERS. NONE CAN LEARN ANYTHING.

The privacy claim is what the papers prove — against three concrete adversaries.

THE NETWORK	Sees only opaque, perfectly-hiding commitments — never amounts, parties, or balances — and cannot link two consecutive transactions of the same token.	PROVEN
THE SENDER	Cannot watch when, or to whom, you later spend what they sent you.	PROVEN
ANYONE WHO KNOWS YOUR ADDRESS	You publish one persistent key; every sender derives a fresh, indistinguishable transaction key against it.	PROVEN

BRIDGES ARE ELIMINATED. THE SOURCE ASSET IS VERIFIED DIRECTLY.

Bridge hacks are the largest loss category in crypto. Unicity has no bridge. A locked source-chain asset is represented as a self-contained token that carries its own state and proof; the recipient verifies the lock directly against the source contract — Unicity does not validate the source chain, recipients do.



NO BRIDGE

There is no cross-chain message to forge and no pooled liquidity to drain.

NO CUSTODIAN

No third party holds the asset between chains; the holder verifies the lock itself.

NOTHING TO HACK

The most dangerous middleman in crypto is structurally gone.

FIVE WINS.
ONE HARD PROBLEM.

Removing the shared ledger buys five things — speed, throughput, privacy, compliance, no bridge — and leaves one: with no shared state, atomicity has to be enforced by the tokens themselves. That is what predicates solve.

THE HARD PART OF CASH-LIKE MONEY IS THE TRADE.

Two parties swap with no one in the middle to hold both legs. The old fix is a timed lock, and the clock is the attack surface — one side stalls and walks. Unicity drops the clock: both commit independently, and the swap forms for both at once or never. This is the atom that turns bearer tokens into a market.

HASHED TIMELOCK (HTLC)

1 · A locks, reveals hash y

2 · B locks against same y

3 · A claims, revealing preimage x

4 · B must claim before timeout

THE TIMING TRAP

B late on step 4 — a dropped connection — and A keeps both.

UNICITY PREDICATE SWAP

A COMMITS
locks their token

B COMMITS
locks their token

UNICITY SERVICE
the shared reference

AUTOMATIC FAIRNESS

- ✓ both locked → swap completes
- ✓ either walked away → both refunded
- ✓ no deadlines, no chasing, done

NO CLOCK

Both commit independently against a shared reference: no deadline, no refund timer.

BOTH OR NEITHER

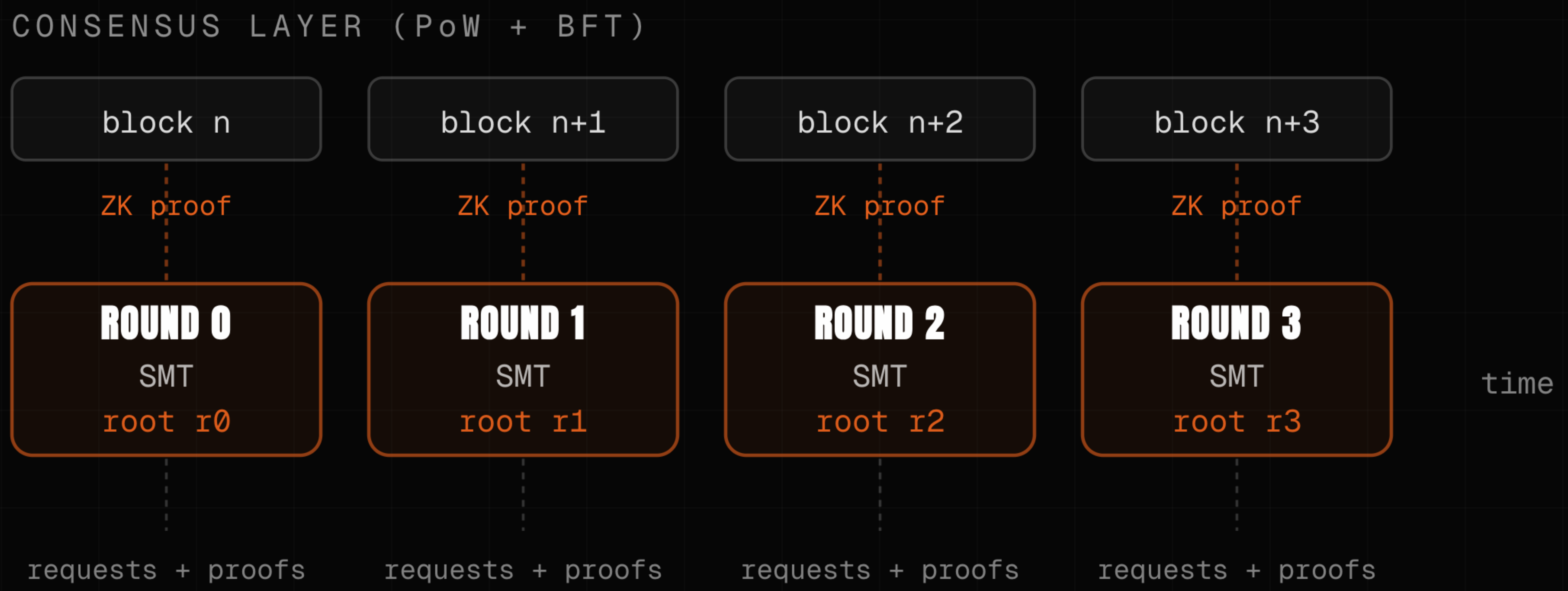
Both locked, the swap completes; either walks away, both keep their token. Capital is never trapped.

NO MEMPOOL, NO MEV

It runs off-chain at machine speed, immune to front-running and mempool games.

THE ORACLE NEVER VALIDATES TRANSACTIONS. THROUGHPUT SCALES BY ADDING SHARDS.

Because the oracle re-executes nothing — it certifies one consistency proof per round, not each payment — cost per transaction collapses and capacity grows by adding shards, not by raising a limit. The oracle is itself permissionless and redundant, so it is not a new central chokepoint.



~30,000 / SEC PER SHARD

A design figure on a single consumer CPU — Plonky3 AIR, no trusted setup. Stated, not a live benchmark.

SUB-MICROCENT PER PROOF

Each new shard adds capacity; the proofs stay succinct.

PERMISSIONLESS, REDUNDANT

Independent staked aggregators join freely; sub-trees run in parallel, with no global bottleneck.

CEX SPEED. DEX CUSTODY. PRIVATE AND COMPLIANT.

When two agents settle directly, the venue in the middle has nothing left to do. Binance has the speed. Uniswap has the custody. Neither has the privacy. Unicity has all three.

	BINANCE	UNISWAP	HYPERLIQUID	UNICITY
SPEED	sub-second	block time	fast	sub-second
SELF-CUSTODY	custodial	you hold keys	you hold keys	you hold keys
PRIVACY	operator sees all	fully public	public positions	unlinkable
MEV	operator order	sandwiched	mempool	no mempool
COMPLIANCE	off-chain KYC	none	none	in the asset

A WHOLE COMPANY, RUN BY A SWARM OF AGENTS.

This is not a roadmap. We built a decentralized autonomous corporation for BlackRock — a fully autonomous parametric, weather-based insurer: capital, underwriting and cession run by agents transacting on Unicity tokens bound to conditional-ownership rules. Agents are the new smart contracts — verifiable code, off-chain, operating on bearer assets.

CAPITAL

Capacity is committed and tracked on-chain-proven bearer tokens.

UNDERWRITING

Premiums priced and bound by predicate to the policy's receive-condition.

CESSION

Claims and reinsurance settle agent-to-agent, with no venue in the middle.

THE SERIOUS PLAYERS ARE SPENDING BILLIONS TO ESCAPE THE LEDGERS THEY BUILT.

An independent analysis of Allaire (Circle), Yakovenko (Solana), Hoskinson (Cardano) and Greg Kidd found unanimous agreement on the coming machine-driven liquidity shock. They optimise the ledger. Unicity removes it.

PLAYER	THEIR MOVE	APPROACH
CIRCLE	Pivoted to infrastructure — the Arc L1, a $222M presale, bought Malachite.	optimise the ledger
SOLANA	Alpenglow cut finality to ~150ms — and its own co-founder concedes a physical ceiling.	optimise the ledger
CARDANO	Midnight adds zero-knowledge selective-disclosure privacy.	optimise the ledger
USBC · GREG KIDD	Put identity inside the dollar — the right conclusion, on one charter.	the right idea
UNICITY	Removes the shared ledger at the data-structure level.	eliminate it

WE BUILT THIS ONCE BEFORE. NOW MAKE IT HOLD BETWEEN STRANGERS.

THE TEAM

Founded Guardtime, built KSI — ledgerless verification for the Estonian e-state, the US DoD, DARPA and NATO. In production since 2012, eIDAS-grade. KSI held **300,000 tx/sec** in Eesti Pank's 2021 test — the lineage, not a live Unicity number.

THE PROOF

Three math papers prove only privacy and no-double-spend — drop them in any model and it checks. Throughput is a sharding design result, named as such. All public on github.com/unicitynetwork.

THE ASK

$5M Seed · $25M cap / $50M token FDV · SAFE + token warrant. First build: a USBC dollar that settles the moment the credential checks out — you supply the regulatory layer we openly lack.

A SINGLE CHARTER SETTLES AT HOME. THIS IS THE LAYER THAT CARRIES THE SAME RULE BETWEEN TWO BANKS, OR TWO MACHINES, THAT SHARE NOTHING ELSE.